NATIONAL 5 ENGINEERING SCIENCE · MECHANISMS AND STRUCTURES

Structures and Forces

Course notes

Version: 26 October 2026

Name _____ **Class** _____

Learning outcomes

Tick each box when you feel confident you can do it. Come back to this page before your unit test.

BY THE END OF THIS TOPIC…	✓
1 I can identify tension, compression, bending, shear and torsion, and give examples of each.	☐
2 I can draw a free-body diagram showing the forces acting on a structure.	☐
3 I can identify the fulcrum, load and effort in a lever. *(not examined)*	☐
4 I can state the principle of moments, and calculate moments and torque.	☐
5 I can calculate an unknown force or distance using the principle of moments.	☐
6 I can calculate the reaction forces at the supports of a beam.	☐
7 I can state the conditions for a structure to be in equilibrium.	☐
8 I can find the size and direction of an unknown force using a scale drawing (triangle of forces).	☐

Where to write your answers: if you have access to a device, use the online course notes — you can type your answers into the answer panels and they are saved as you go. Otherwise, write your answers in your jotter, with the task and question number beside each answer.

Scan for the online course notes

Section 1: Types of Force

Forces impact structures in a variety of different ways depending on how they are applied. Forces can move a structure slightly or cause damage by changing its shape. Sometimes the changes are almost impossible to see — for example, a bridge will sag slightly when a vehicle drives over it. Forces can also make an object move or change direction.

> Forces are measured in **Newtons (N)**.

There are five types of force you need to know:



Compression

Forces acting to squash or shorten a structure — pushing inwards from both ends



Tension

Forces acting to stretch or lengthen a structure — pulling outwards from both ends



Bending

Forces acting across the length of a structure causing it to bend or deflect



Shear

Forces acting in opposite directions across a section — like cutting or tearing



Torsion

Forces that twist or turn a structure about its own axis

Compression

The supporting columns of the Queensferry Crossing experience a huge downward force from the weight of the platform and traffic crossing it. The ground exerts an upward reaction force on the bases of the columns. These two opposing forces try to squash or shorten the column — the column is said to be in **compression**. A weightlifter's body experiences compression in the same way: the weights push down, the ground pushes up.

Tension

Tension is the opposite of compression. A structure under tension is being **pulled apart**. In a tug of war, both teams pull the rope in opposite directions — the rope is in tension. Wire ropes, cables and chains are typically in tension.

Bending

Structures that carry loads across their length are subject to **bending forces**. A weightlifting bar bends due to the downward forces of the weights at each end. Beams in buildings and bridges experience bending.

Shear

Shear forces can be described as tearing or cutting forces. A pair of scissors cuts a ribbon using shear — the two blades push in opposite directions across the material. A lawn mower blade shears the grass.

Torsion

Torsion (also called torque) forces twist or turn a structure. A screwdriver being twisted to drive a screw applies torsion to the screw. A spanner turning a bolt applies torsion to the bolt.

✏️ Task 1 — Types of force

For each of the five types of force, write down two real-life examples where that force is present.

Section 2: Free-Body Diagrams

Before working through levers, moments and beams, it helps to have one tool that will be used again and again: the **free-body diagram**. It is simply a simplified sketch that shows all the forces acting on a structure or body, with everything else stripped away. It isolates the object from its surroundings and replaces real components (hinges, supports, ropes, the ground) with arrows representing the forces they exert.

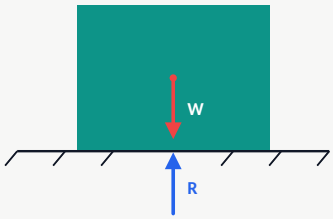
To draw a free-body diagram:

1. Draw a simple outline of the structure (or just a line or box)
2. Replace all connections and supports with arrows showing the force they exert
3. Label each force with a letter or value
4. Show downward forces (loads, weight) acting through their point of application
5. Show upward reaction forces at support points

> Free-body diagrams make it easier to see and calculate all the forces acting on a structure. Every force must be accounted for — including the weight of the structure itself if stated.

WORKED EXAMPLE — A CRATE RESTING ON THE GROUND

A crate sits still on flat ground. Two forces act on it: its own **weight** (W), pulling straight down, and the **reaction** from the ground (R), pushing straight up. Because the crate is not moving, these two forces must be equal and opposite.



1. **Isolate the crate** and remove the ground — draw a box in its place
2. **Add the weight** as an arrow pointing straight down from the middle of the crate, labelled W
3. **Add the reaction** as an arrow pointing straight up from the base of the crate, labelled R
4. Because the crate is staying still, **R = W** — the two forces balance

Mass and weight are not the same thing

Mass (m)

- The amount of **matter** in an object
- Measured in **kilograms (kg)**
- Stays the **same everywhere**: on Earth, on the Moon, or in space

Weight (W)

- The **force** of gravity pulling an object down
- Measured in **newtons (N)**, like every other force
- **Changes** with gravity: the same object weighs less on the Moon

$$\text{Weight} = \text{mass} \times g \qquad W = m \times 9.8$$

On Earth, g = 9.8 m/s², so every kilogram of mass has a weight of about 9.8 N. For example, a 50 kg crate has a weight of 50 × 9.8 = **490 N**.

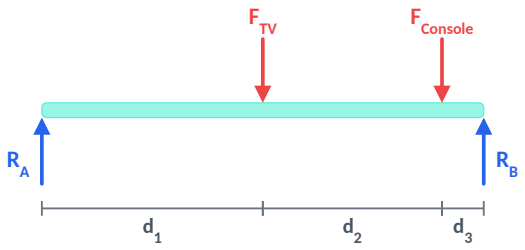
In everyday speech people say “I weigh 60 kg”, but that is really their **mass**. On a free-body diagram, always draw **weight**, a force in newtons, never mass in kilograms.

WORKED EXAMPLE — A TV AND GAMES CONSOLE ON A TABLE

Think of a free-body diagram as a simplified way of seeing a structure. When you look at a TV sitting on a table, you see the wood, the screen and the cables. Put on your **engineering glasses** and all of those physical details disappear, leaving only a skeleton of the hidden pushes and pulls that keep everything perfectly still.



What you see



What an engineer sees

In the free-body diagram, the table top becomes a single beam. The weight of the TV (F_{TV}) and the weight of the games console ($F_{Console}$) push **down** on it, exactly where each one sits.

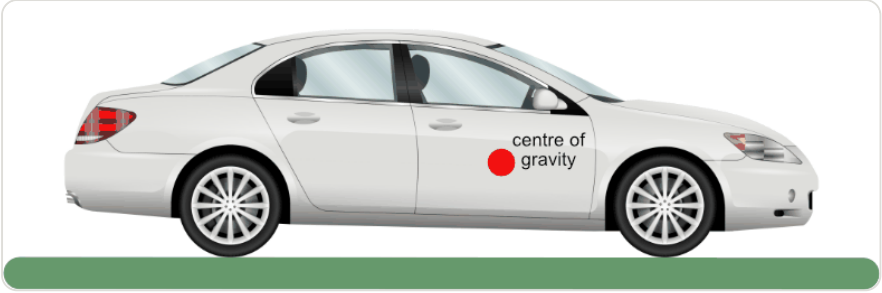
The two table legs **react against** those downward forces by pushing **up** on the table top. These upward forces are the reactions R_A and R_B. The table is not moving, so the legs must push up with exactly as much force as the TV and console push down:

$$R_A + R_B = F_{TV} + F_{Console}$$

The legs don't share the load equally. The TV sits in the middle, so each leg carries about half of its weight, but the console sits much closer to leg B, so leg B carries most of that. The distances d_1, d_2 and d_3 are what let you work out exactly how much each leg carries, using moments in Section 5.

✏️ Task 2 — Free-body diagrams

1. Draw a free-body diagram of a car resting on flat ground. Show the downward forces (weight acting through the centre of gravity, also known as the centre of mass) and the reaction forces at each axle. Use suitable letter labels.



2. A boy balances on one foot on a wooden beam which rests on a barrel at each end. Ignoring the weight of the beam, draw and label a free-body diagram of the beam showing all forces acting on it. Use suitable letter labels.



Section 3: Levers

> **Note:** levers are not examined in National 5 Engineering Science. They are included here because they are a useful introduction to the next section, **Moments**.

A lever is a simple machine that can **amplify force**. It is made up of a rigid bar that turns about a pivot point called the **fulcrum**. Forces act on the lever at different points along its length, and the lever's job is to turn one of those forces into a bigger (or more useful) one somewhere else.

> **Fulcrum** = the pivot point that the lever turns about. Every lever has a fulcrum and at least two forces acting on it, on either side of, or at different distances from, that fulcrum.

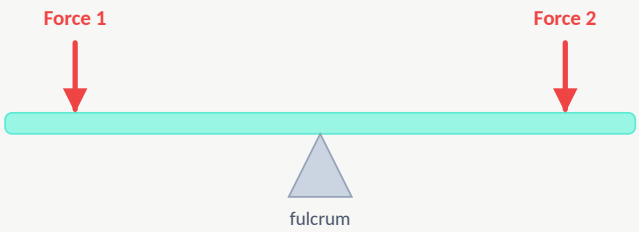
You already know from free-body diagrams that every force acts **at a specific point**. With a lever, what matters most is not what each force is called, but **where it acts relative to the fulcrum**. Moving the fulcrum, or moving a force closer to or further from it, changes how the lever behaves — this is the idea you'll use again in the next section on moments.

The fulcrum can sit in different places

The fulcrum does not have to be positioned halfway between the two forces. Depending on where it sits, a lever can behave very differently. Here are three arrangements you will meet often — pay attention to the **order** the fulcrum and the two forces appear in, left to right.

ARRANGEMENT 1 — FULCRUM BETWEEN THE TWO FORCES

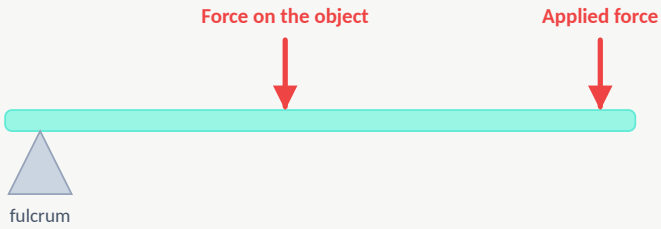
The fulcrum sits in the middle, with one force acting on each side of it.



Examples: a **seesaw** (a child sits at each end, the pivot block is in the middle); a pair of **scissors** (the pivot screw is between the finger holes and the blades); a **crowbar** resting over a fixed edge, lifting something on the other side.

ARRANGEMENT 2 — FULCRUM AT ONE END, ONE FORCE IN THE MIDDLE

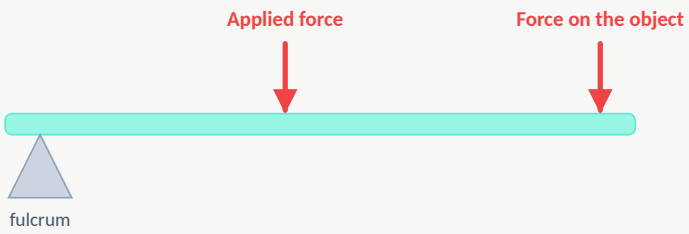
The fulcrum sits at one end. The force being moved acts somewhere in the middle, and the force you apply acts at the far end.



Examples: a **wheelbarrow** (fulcrum at the wheel, the weight of what's inside acts partway along, your hands apply force at the handles, furthest from the wheel); a **bottle opener** (fulcrum at the lip resting on the bottle rim, the force lifting the cap acts in the middle, your hand applies force at the far end); a **nutcracker**.

ARRANGEMENT 3 — FULCRUM AT ONE END, THE APPLIED FORCE IN THE MIDDLE

The fulcrum sits at one end. This time, the force you apply acts somewhere in the middle, and the force being moved acts at the far end.



Examples: a **fishing rod** (fulcrum at the hand nearest the reel, the other hand applies force further up the rod, the fish pulls at the tip — the furthest point away); a pair of **tweezers**; a **broom** (top hand acts as the fulcrum, the lower hand applies force, the bristles act at the far end).

Levers in everyday objects

A **door** is also a lever — the hinge is the fulcrum. Push near the hinge and the door is hard to move; push at the handle, as far from the hinge as possible, and it swings open easily with far less force. This is exactly the same idea you'll use in the next section: the further a force acts from the fulcrum, the bigger its turning effect.

> You do not need to remember special names for these arrangements. What matters is being able to look at a lever — in a diagram or in a photo — and identify: where the fulcrum is, and where each force acts.

When a counterbalance weight is attached to one side of a lever to assist the user, it is known as a **counterbalance**. Cranes and diggers use counterbalances to prevent them toppling over.

✏️ Task 3 — Identifying the fulcrum

Examine the photos below to identify: (a) where the fulcrum is, (b) where each force acts, and (c) which arrangement it matches — fulcrum in the middle, or fulcrum at one end with the object being moved or the applied force in the middle.



1. Door handle

2. Seesaw

3. Wheelbarrow

4. Fishing rod

Section 4: Moments

A force does not always make an object move in a straight line — very often it makes the object **turn** instead. The turning effect of a force about a point is called a **moment**. You will also meet the word **torque**: torque is a turning force, and it is the word engineers usually use when something is twisted about its own axis, such as a spanner turning a nut or a motor turning a shaft. A moment and a torque are worked out in exactly the same way (force × distance). Every moment acts around a fixed point called the **pivot** (or fulcrum): the point about which the object is free to rotate. On a see-saw, the pivot is the block in the middle. On a spanner tightening a bolt, the pivot is the bolt itself. On a door, the pivot is the hinge. Moments and torque are both measured in **newton-metres (Nm)**.

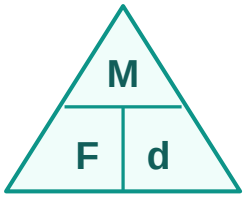
> The size of a moment depends on two things: **the size of the force** applied, and **the perpendicular distance** between the pivot and the line along which the force acts. This distance must always be measured at right angles to the force — not along the beam or lever itself.

Moment or Torque (Nm) = Force (N) × Distance (m)

Distance is measured at right angles from the line of action of the force to the pivot

The MFd triangle. Cover up the quantity you want to find, and what is left shows you how to work it out.

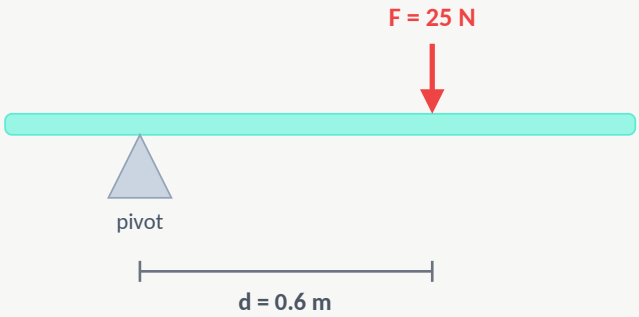
- Cover **M**: F and d sit side by side, so **M = F × d**



- Cover **F**: M sits over d, so **F** = $\frac{M}{d}$
- Cover **d**: M sits over F, so **d** = $\frac{M}{F}$

WORKED EXAMPLE — CALCULATING A SINGLE MOMENT

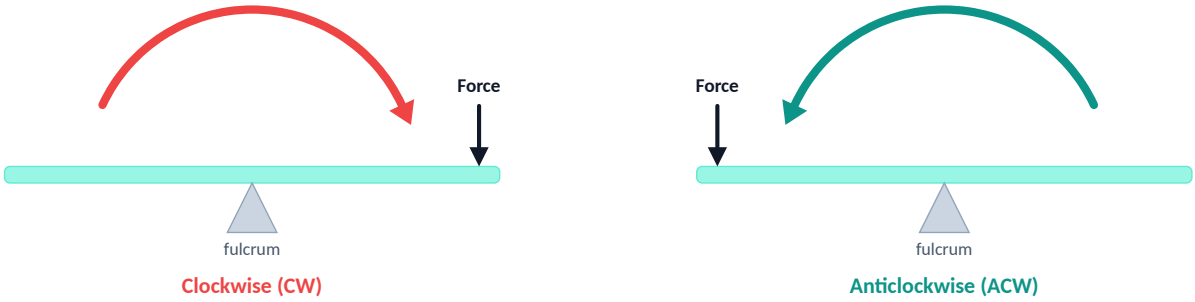
To open a heavy garden gate, a person pushes on the edge furthest from the hinge with a force of 25 N. The hinge (the pivot) is 0.6 m from where the force is applied, measured perpendicular to the push. Calculate the moment used to open the gate.



1. **Write down what you know:** F = 25 N, d = 0.6 m, M = ?
2. **Apply the formula:** M = F × d
3. M = 25 × 0.6 = **15 Nm**

Equilibrium and the Principle of Moments

When a system is in a steady, balanced state it is said to be in **equilibrium**. For a lever or beam to be in equilibrium, the clockwise moments must equal the anticlockwise moments — one side has no more turning effect than the other, so nothing rotates.



Sometimes there is more than one force turning a beam in the same direction — for example, two children sitting on the same side of a see-saw. Rather than writing out every individual moment, we use the symbol **∑** (the Greek capital letter **sigma**). In maths and science, ∑ simply means **"add up all of the following."** So ∑CWM means: add up every clockwise moment acting on the beam, however many there are, to get one total.

In words, the **Principle of Moments** says:

> The sum of the clockwise moments equals the sum of the anticlockwise moments.

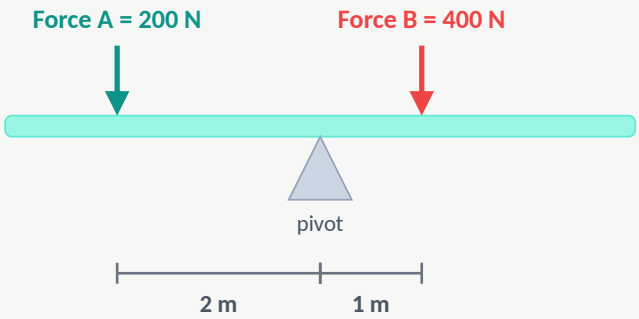
$$\sum CWM = \sum ACWM$$

$F_1 \times d_1 = F_2 \times d_2 \ldots$

(the dots show that if more than one force acts on a side, you add all of their moments together first)

WORKED EXAMPLE 1 — IS THE LEVER IN EQUILIBRIUM?

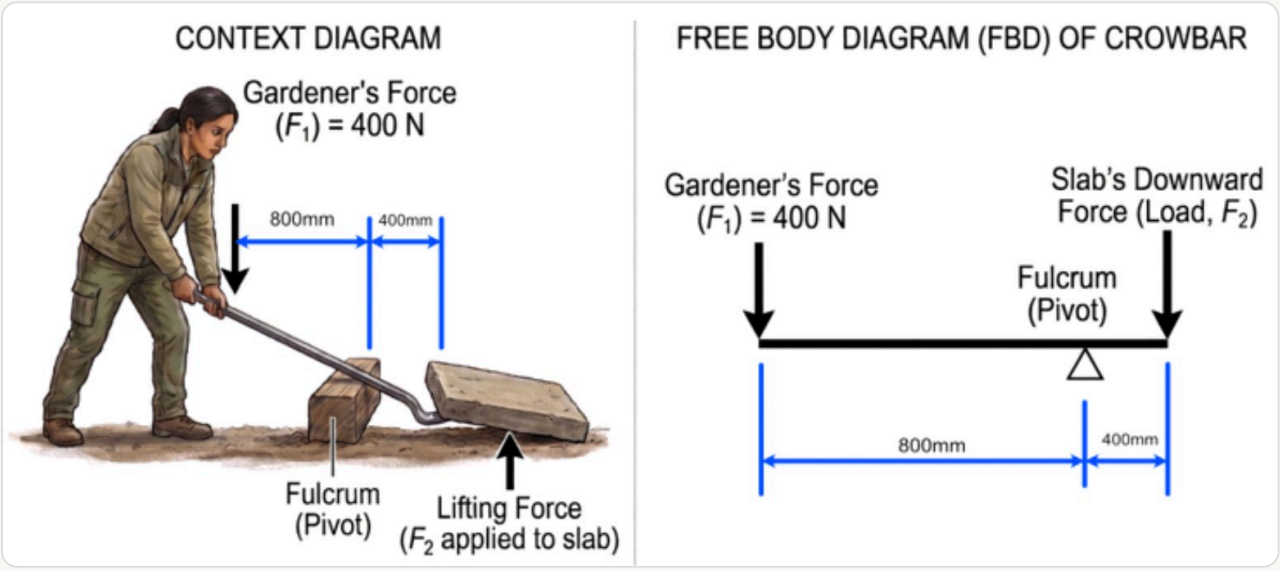
A lever has a force of 400 N acting 1 m from the fulcrum on the right, and a force of 200 N acting 2 m from the fulcrum on the left. Is the lever in equilibrium?



1. **Calculate the clockwise moment:** Force B × distance = 400 N × 1 m = 400 Nm
2. **Calculate the anticlockwise moment:** Force A × distance = 200 N × 2 m = 400 Nm
3. **Compare:** 400 Nm = 400 Nm ⇒ ∑CWM = ∑ACWM ⇒ **The lever is in equilibrium.**

WORKED EXAMPLE 2 — CROWBAR LIFTING A PAVING SLAB

A gardener uses a crowbar and a wooden block to lift a heavy paving slab. She exerts a downward force of 400 N, 800 mm from the fulcrum. The slab is 400 mm from the fulcrum on the other side. Calculate the force of the slab on the end of the crowbar when all the forces are in equilibrium.

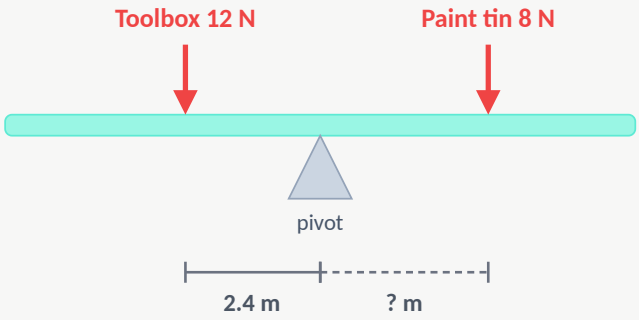


1. ∑ Moments = 0, so **∑ Clockwise Moments = ∑ Anticlockwise Moments**
2. $(F_{slab} \times 400) = (400 \times 800)$
3. $F_{slab} = \frac{400 \times 800}{400}$ = **800 N**

The distances can stay in millimetres here, because both sides of the equation use the same unit.

WORKED EXAMPLE 3 — FINDING AN UNKNOWN DISTANCE

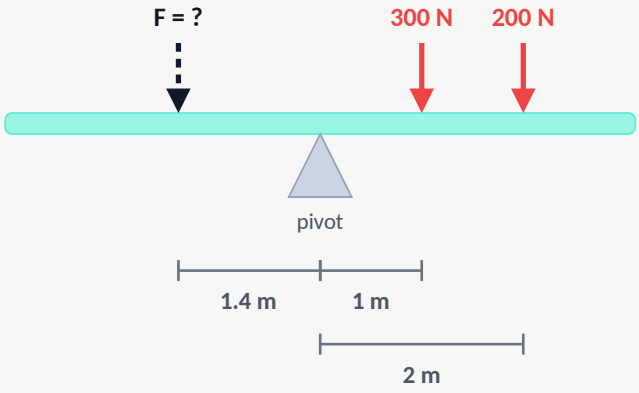
A wooden plank is balanced across a fixed support so that it can pivot freely. A toolbox weighing 12 N is placed 2.4 m from the pivot. A tin of paint weighing 8 N is placed on the other side. How far from the pivot must the tin of paint be placed for the plank to balance?



1. ∑ Moments = 0, so **∑ Clockwise Moments = ∑ Anticlockwise Moments**
2. $(8 \times d) = (12 \times 2.4)$
3. $8 \times d = 28.8$
4. $d = \frac{28.8}{8}$ = **3.6 m** — the paint tin must be placed 3.6 m from the pivot.

WORKED EXAMPLE 4 — BALANCING SEVERAL FORCES

A see-saw is pivoted at its centre. On the right-hand side, one child applies a downward force of 300 N at 1 m from the pivot, and a second child applies 200 N at 2 m from the pivot. What downward force must an adult apply at 1.4 m from the pivot, on the left-hand side, to balance the see-saw?



1. ∑ Moments = 0, so **∑ Clockwise Moments = ∑ Anticlockwise Moments**
2. $(300 \times 1) + (200 \times 2) = (F \times 1.4)$
3. $700 = F \times 1.4$
4. $F = \frac{700}{1.4}$ = **500 N**

✏️ Task 4 — Moments calculations

Use the Principle of Moments to find the missing force or distance in each problem. Show all working.

1. A lever has a force of 300 N applied 1.5 m from the fulcrum. A second force acts 0.5 m from the fulcrum on the other side. Calculate the size of the second force needed for equilibrium.

2. A downward force of 600 N acts 2 m from a fulcrum. Where must a force of 400 N be applied on the other side to balance the lever?

3. A lever has two forces on the right of the fulcrum: 200 N at 3 m and 400 N at 1.5 m. A force is applied 2 m to the left of the fulcrum. Calculate the force needed for equilibrium.

4. A spanner is used to tighten a bolt. A force of 150 N is applied to the end of the handle, 0.25 m from the centre of the bolt. Calculate the torque applied to the bolt.

5. A see-saw is balanced with a pivot at its centre. A child sits 1.8 m from the pivot and applies a downward force of 120 N. Calculate the distance from the pivot at which a force of 90 N must be applied on the other side to keep the see-saw balanced.

Section 5: Beams and Reactions

A beam is a structural member that carries loads across its length. For a horizontal beam to be in **equilibrium**, two conditions must be satisfied simultaneously:

$$\sum \text{upward forces} = \sum \text{downward forces}$$

$$\sum \text{clockwise moments} = \sum \text{anticlockwise moments}$$

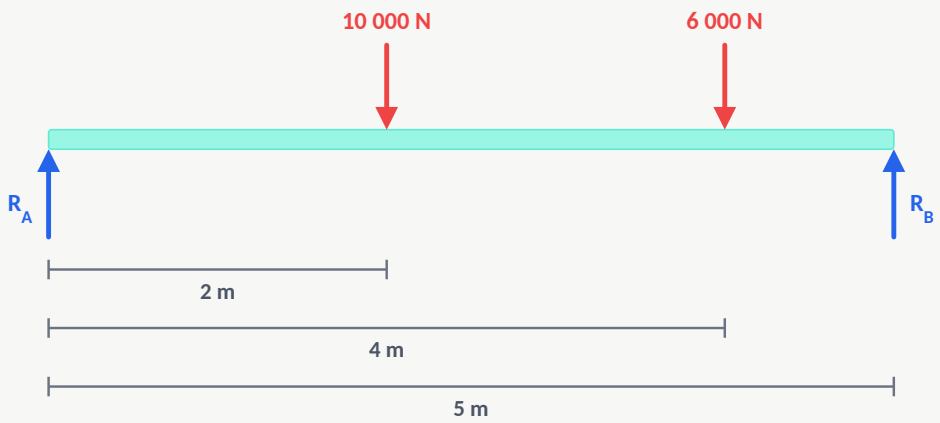
The upward forces at the support points are called **reactions** (R_A and R_B). To find both reactions, we use the Principle of Moments and the force balance equation together. A free-body diagram of the beam — showing every downward force and both reactions — is the best starting point for any of these problems.

Method for finding beam reactions

1. **Take moments about one reaction** (R_A) — this eliminates R_A from the equation, allowing you to solve for R_B
2. **Use the force balance** ($\sum$ up = $\sum$ down) to find R_A
3. **Check:** $R_A + R_B$ should equal the sum of all downward forces

WORKED EXAMPLE — SIMPLY SUPPORTED BEAM WITH TWO LOADS

A 5 m beam is supported at each end (R_A at the left, R_B at the right). Two downward loads are applied: 10 000 N at 2 m from R_A, and 6 000 N at 4 m from R_A. Beam weight is ignored. The free-body diagram below shows every force acting on the beam, with all distances measured from R_A.



1. **Take moments about R_A** (so R_A drops out):
   ∑CWM = ∑ACWM
   $(10\,000 \times 2) + (6\,000 \times 4) = R_B \times 5$

2. $20\,000 + 24\,000 = R_B \times 5$
   $44\,000 = R_B \times 5$
   R_B = 8 800 N

3. **Use force balance:**
   $R_A + R_B = 10\,000 + 6\,000 = 16\,000$ N
   $R_A = 16\,000 - 8\,800 =$ **7 200 N**

4. **Check:** 7 200 + 8 800 = 16 000 N ✓

Checking your answers with the NoStress simulator

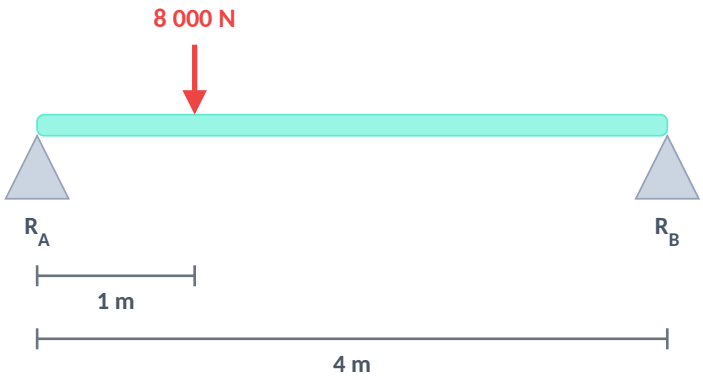
NoStress is a beam simulator. You build a beam, add its supports and loads, then press **Solve** to see the reactions at each support. **Show Working** then sets out the moments step by step, so you can compare the simulator's method with your own. Use it to **check** answers you have already worked out, not to replace the working: in an assessment you need to show every step yourself.

Try it: the two-load beam from the worked example
This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

✏️ Task 5 — Beam reactions

Show all working for each problem. Remember: take moments about one reaction first, then use the force balance to find the other.

1. A 4 m beam is simply supported at each end, R_A at the left and R_B at the right. A single load of 8 000 N acts 1 m from R_A. Determine both reactions.



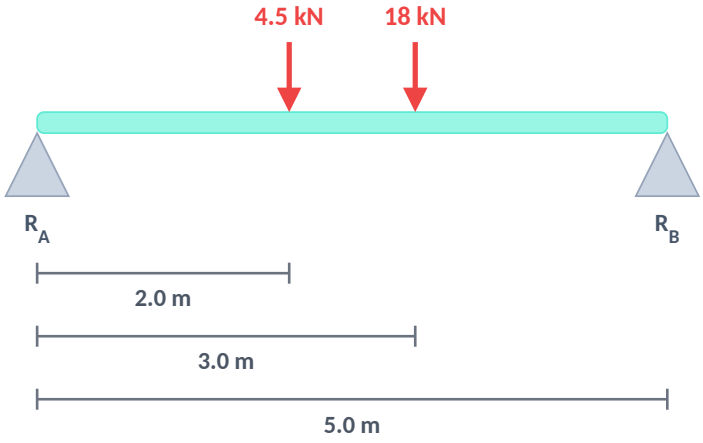
2. A 6 m beam is supported at each end. Loads of 4 000 N at 1.5 m and 7 000 N at 4 m from the left support act downwards. Determine both reactions.

3. A 7 m horizontal beam is part of a bridge structure, simply supported at A (left) and D (right). Loads of 6 000 N at 2 m, 3 000 N at 3.5 m, and 9 000 N at 5.5 m from A act downward. Determine the reactions at A and D.

4. A car has a total weight of 12 000 N. The front axle (R_F) is 0.8 m from the front and the rear axle (R_R) is 2.7 m from the front. The centre of mass acts 1.2 m from the front. Calculate both axle reactions.

5. A contestant in a weight-lifting competition performs a barrel lift. The barrel weighs 800 N and is positioned 1.2 m from the pivot. The contestant's lifting force F acts 0.4 m from the pivot on the other side. Calculate the lifting force F.

6. A road sign gantry spans 5.0 m between its two support posts, R_A at the left and R_B at the right. A sign produces a downward force of 4.5 kN, acting 2.0 m from R_A. The gantry's own weight of 18 kN acts at 3.0 m from R_A. Calculate the reaction force R_B by taking moments about R_A.



Check your Task 5 answers
This interactive simulator is in the online course notes. Scan the QR code, or use a device, to open it.

📖 Once you can calculate the forces in structural members, the next step is to check whether the **material** chosen can withstand those forces. Stress and strain calculations are covered in the **Materials** course notes.

Section 6: Vectors and Equilibrium

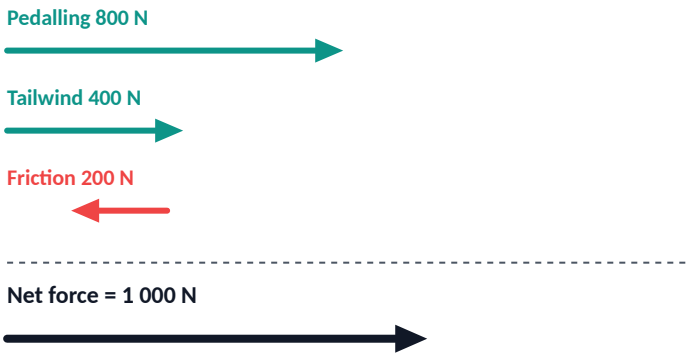
Force is a **vector quantity** — it has both **magnitude** (size) and **direction**. Vectors can be represented as drawn lines where:

- The **length** of the line (drawn to scale) represents the magnitude of the force
- The **direction** and arrowhead show the direction of the force

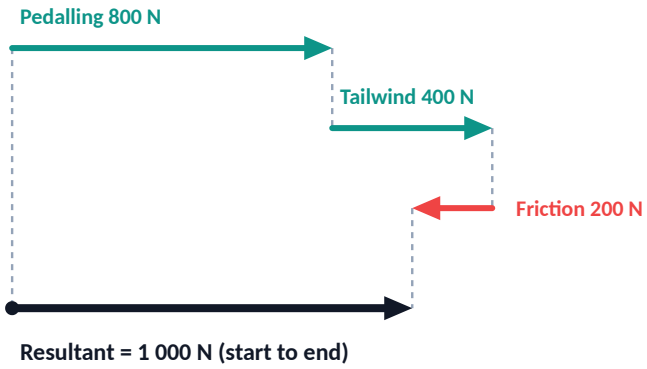


Drawn to the same scale, the 600 N vector is exactly twice the length of the 300 N vector.

When multiple forces act along the same line, they can simply be added or subtracted. For example: a cyclist pedalling with 800 N, assisted by a tailwind of 400 N, but with 200 N road friction, has a net force of 800 + 400 − 200 = **1 000 N**.



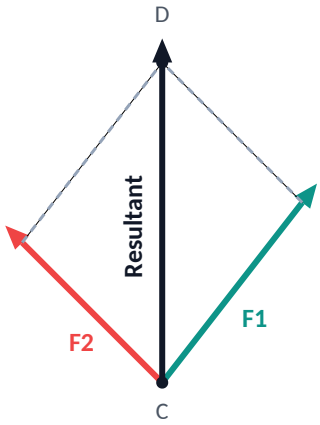
The same forces can also be added by drawing them **tip to tail**: draw the first vector, then start the next one from the tip (arrowhead) of the one before, and so on. The **resultant** is a single arrow drawn from the **very start** of the first vector to the **very end** of the last one. This works for any number of forces, and it is the same idea used for angled forces below.



Resultant of two angled forces

When two forces act at an angle to each other, their combined effect is a single force called the **resultant**. The resultant can be found graphically using the **parallelogram of forces**:

1. Choose a suitable scale and draw both force vectors from a common point C
2. From the tip of each vector, draw a line parallel to the other vector
3. The point where these lines intersect (D) completes the parallelogram
4. Draw the line from C to D — this is the **resultant**
5. Measure the length of C to D and convert using your scale to find the magnitude
6. Measure the angle to find the direction



> The **equilibrant** is the single force that would exactly balance the resultant — it is equal in magnitude but opposite in direction to the resultant.

Conditions for Equilibrium

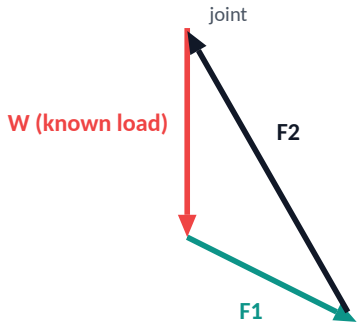
For a structure or body to be in **equilibrium** (a balanced, stable state), three conditions must all be satisfied:

Condition	Meaning
∑ upward forces = ∑ downward forces	Vertical forces balance — no net vertical movement
∑ leftward forces = ∑ rightward forces	Horizontal forces balance — no net horizontal movement
∑ CWM = ∑ ACWM	Moments balance — no net rotation

Graphical solution for forces in members

For a structure such as a crane or frame, the forces in individual members can be found graphically by drawing a **triangle of forces**:

1. Identify the joint where the known load acts
2. Draw the known force to scale
3. From each end of this line, draw lines parallel to the unknown force members
4. Where the lines intersect completes the triangle
5. Measure each side to find the forces (following arrowheads round the triangle indicates compression or tension)



✏️ Task 6 — Vectors and equilibrium

1. State whether each of the following is TRUE or FALSE and explain your reasoning:

   a) A body that is accelerating is in a state of equilibrium.

   b) For a body to be in equilibrium it is necessary only for the vector sum of forces to be zero.

   c) A resultant force is a single force that can replace two or more forces.

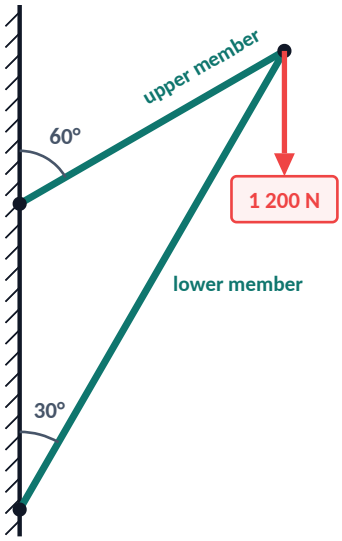
   d) If two or more forces are replaced by a resultant force, the effect on the body is changed.

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2. Describe the three conditions necessary for a structure or body to be in equilibrium.

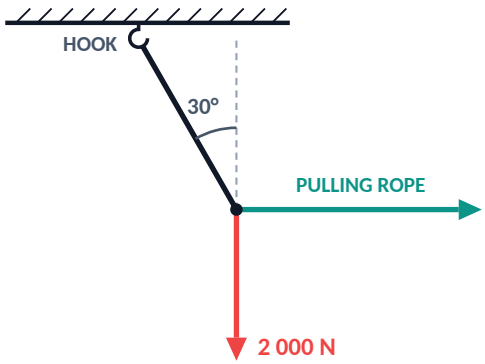
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3. A small crane on a fishing trawler is used to lift cases of fish on to the dock. The crane is fixed to a vertical wall, with its upper member at 60° to the wall and its lower member at 30° to the wall. The load being lifted weighs 1 200 N. Using a scale of 10 mm = 200 N, draw a triangle of forces to find the size of the force in each crane member, and state whether each member is in tension or compression.



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4. A weight of 2 000 N hangs from a rope tied to a hook that is fixed firmly to a roof joist. A second rope is tied to the hanging rope and pulled horizontally until the upper part of the hanging rope sits at 30° to the vertical. Using a suitable scale (for example 10 mm = 200 N), draw a triangle of forces to find the horizontal pull on the second rope and the force on the hook.



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5. The diagram shows a cranked lever from a gear-change mechanism. A horizontal force of 750 N acts at the top pin and a vertical force of 600 N acts at the right-hand pin. Using a scale of 10 mm = 100 N, draw the two forces tip to tail to find the resultant force F_R acting on the hinge pivot and the angle θ it makes with the horizontal.

